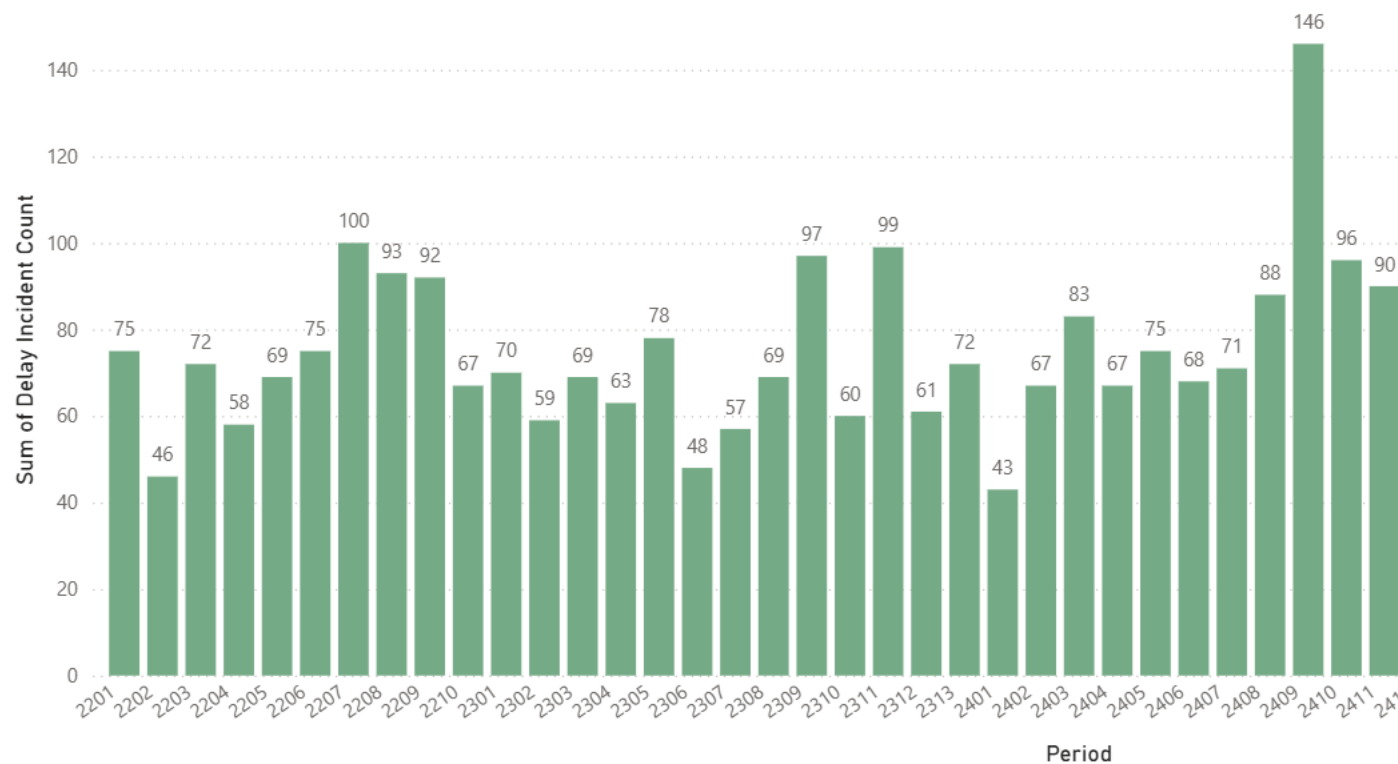


Figure 1

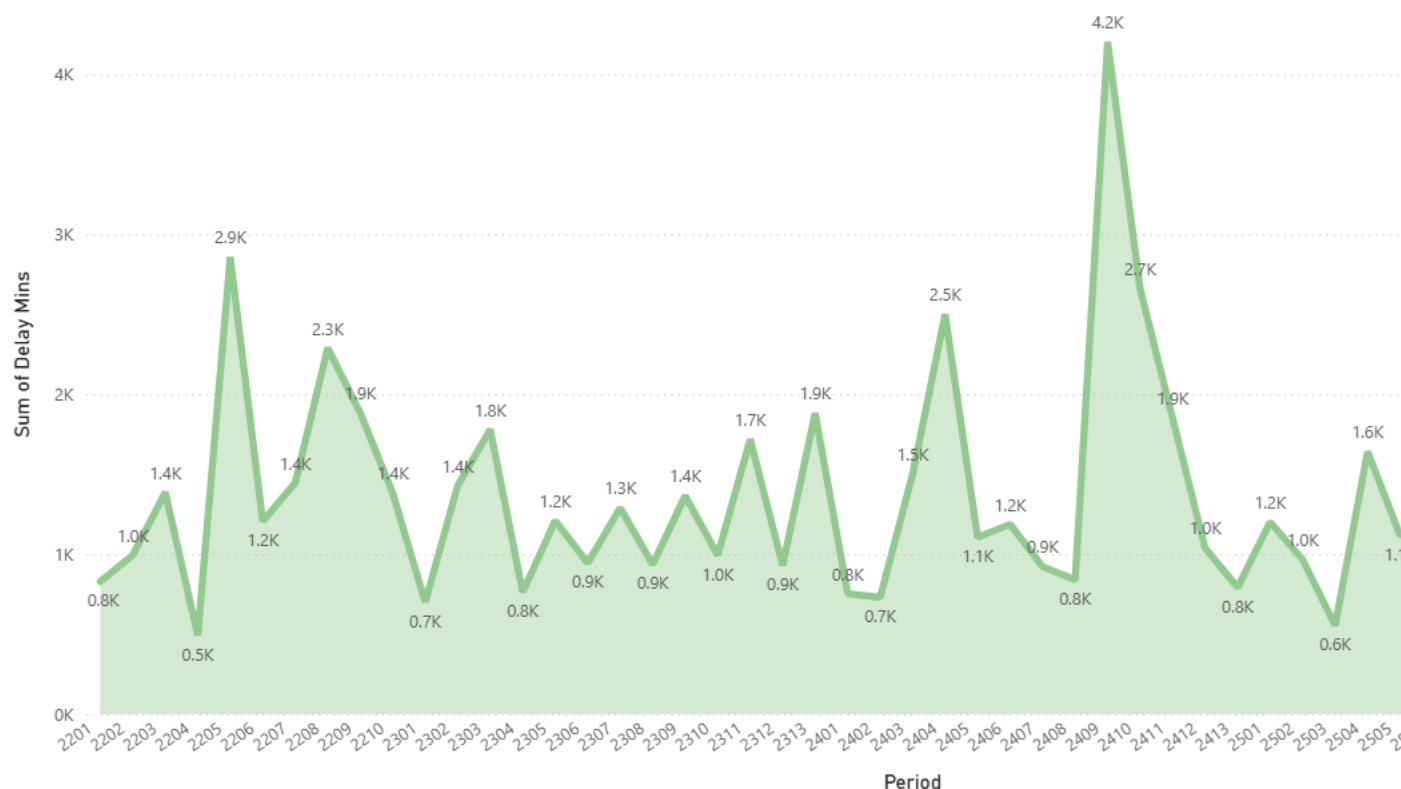
Fleet Delay Incidents By Period (P2201-P2611)



The clustered column chart above shows the number of fleet delay incidents recorded across the operational periods P2201 to P2611. Each column represents the total number of delays within a specific operational period. Therefore, comparisons can be made between operational reliability over time. The chart implies that most periods experience between 60 to 100 incidents each period. However, there are noticeable spikes. For instance, in period 2409, there was a steep increased compared to period 2408, where P2408 had 88 incidents in comparison to 146 in Period 2409. This indicates that during this period, there was increased disruption which resulted in fleet delays. It is essential for LNER to identify periods with higher delay frequencies. This is because repeated disruptions may potentially impact fleet availability and increase the risk of timetable instability.

Figure 2

Fleet Delay (minutes) by Operational Period (P2201-P2611)



The line chart above shows the trend in fleet delay minutes across operational periods P2201 to P2611. Each point in the chart represents the sum of delay minutes which were recorded within a specific operational period. Having this line chart, comparisons between each data point from a particular operational period can be observed. While most periods show a sum delay time of around 800 to 1500 minutes, noticeable spikes can be seen. For example, the total delay time across period 2410 exceeded 4000 minutes compared to period 2409 where the total delay time was only around 800 minutes. Spikes in the chart signifies periods of high operational disruption. These spikes could stem due to reliability issue, operational constraints, or faults within LNER's infrastructure. It is very important for LNER to analyse the trends between delay times across time because the increases could signal more pressure on fleet availability and higher risks of service disruption.

Figure 3

Operational Issues by Late Started Events (Depot)

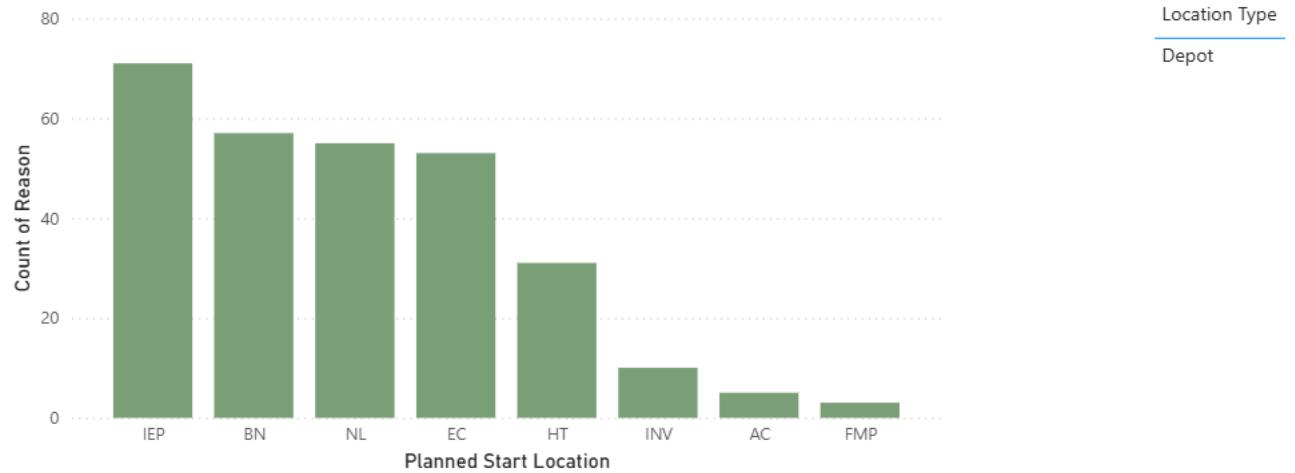


Figure 3 shows a clustered column chart which shows a depot-level breakdown of operational issues. These issues result in late fleet starts from the start location. Each column in the chart represents the total number of issues between each depot. Looking at the chart, we can see that IEP, BN, NL, and EC have the highest incident counts. Whereas, HT, INV, AC, and FMP show lower incident counts. This implies that the pressure for fleets to be rolled out may be limited to certain key maintenance location. LNER can track which depots have the most frequent incidents counts. This can help in solving how to improve fleet availability and reducing service disruptions since operational improvements and maintenance interventions can be prioritised depending on location.